

CHAPTER 1

Introduction

THE BIRTH AND DEVELOPMENT OF GENETICS AND GENOMICS

Few areas of science and medicine are seeing advances at the pace we are experiencing in the related fields of genetics and genomics. It may appear surprising to many students today, then, to learn that an appreciation of the role of genetics in medicine dates back well over a century, to the recognition by the British physician Archibald Garrod and others that Mendel's laws of inheritance could explain the recurrence of certain clinical disorders in families. During the ensuing years, with developments in cellular and molecular biology, the field of **medical genetics** grew from a small clinical subspecialty concerned with a few rare hereditary disorders to a recognized medical specialty whose concepts and approaches are important components of the diagnosis and management of many disorders, both common and rare.

At the beginning of the 21st century, the **Human Genome Project** provided a virtually complete sequence of human DNA—our **genome** (the suffix *-ome* coming from the Greek for "all" or "complete")—which now serves as the foundation of efforts to catalogue all human genes, understand their structure and regulation, determine the extent of variation in these genes in different populations, and uncover how genetic variation contributes to disease. The human genome of any individual can now be studied in its entirety, rather than one gene at a time. These developments are making possible the field of **genomic medicine**, which seeks to apply a large-scale analysis of the human genome and its products, including the control of gene expression, human gene variation, and interactions between genes and the environment, to medical care.

GENETICS AND GENOMICS IN MEDICINE

The Practice of Genetics

The medical geneticist is usually a physician who works as part of a team of health care providers, including many other physicians, nurses, and genetic counselors, to evaluate patients for possible hereditary diseases. They characterize the patient's illness through careful history taking and physical examination, assess possible modes of inheritance, arrange for diagnostic testing, develop treatment and surveillance plans, and participate in outreach to other family members at risk for the disorder.

However, genetic principles and approaches are not restricted to any one medical specialty or subspecialty; they permeate many, and perhaps all, areas of medicine. Here are just a few examples of how genetics and genomics are applied to medicine today:

- A pediatrician evaluates a child with multiple congenital malformations and orders a high-resolution genomic test for submicroscopic chromosomal deletions or duplications that are below the level of resolution of routine chromosome analysis (Case 32).
- A genetic counselor specializing in hereditary breast cancer offers education, testing, interpretation, and support to a young woman with a family history of hereditary breast and ovarian cancer (Case 7).
- An obstetrician sends a chorionic villus sample taken from a 38-year-old pregnant woman to a cytogenetics laboratory for confirmation of abnormalities in the number or structure of the fetal chromosomes, following a positive screening result from a noninvasive prenatal blood test (see Chapter 17).
- A hematologist combines family and medical history with gene testing of a young adult with deep venous thrombosis to assess the benefits and risks of initiating and maintaining anticoagulant therapy (Case 46).
- A surgeon uses gene expression array analysis of a lung tumor sample to determine prognosis and to guide therapeutic decision making (see Chapter 15).
- A pediatric oncologist tests her patients for genetic variations that can predict a good response or an adverse reaction to a chemotherapeutic agent (Case 45).
- A neurologist and genetic counselor provide *APOE* gene testing for Alzheimer disease susceptibility for a woman with a strong family history of the disease so she can make appropriate long-term financial plans (Case 4).
- A forensic pathologist uses databases of genetic polymorphisms in his analysis of DNA samples obtained from victims' personal items and surviving relatives to identify remains from an airline crash.
- A gastroenterologist orders genome sequence analysis for a child with a multiyear history of life-threatening and intractable inflammatory bowel disease. Sequencing reveals a mutation in a previously unsuspected